

Linear Algebra 101 — Part 4

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This is a series of articles towards understanding Linear Algebra. I strongly believe it's important to understand the basics and equations before actually stepping into more advanced topics. Linear Algebra is one of those basics that you have to be really familiar with. Think of it as like, say... programming. There are a bunch of cool libraries out there and you can do cool stuff just by copy and pasting. However, if you want to do something really cool, something no one has done before, you need to know what's behind the scene so that you could code it yourself. Anyways, that's enough for motivation.

I'm following the famous lectures by Dr. Gilbert Strang at MIT for the topics covered in this series.

Video Lectures

Don't show me this again This is one of over 2,200 courses on OCW. Find materials for this course in the pages linked...

ocw.mit.edu

Materials covered in Part 4:

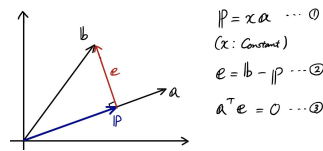
In this story, I'm going to cover the followings. In Part 1–3 we already covered the basics and from here on, the topics will be based on the fundamentals that we already learned.

- Projections
- Orthonormal vectors
- Orthogonal matrix
- Gram Schmidt process

The corresponding lectures from the videos by Dr. Gilbert Strang for these topics are Lectures 15–17 and I would recommend you actually watch his talks for better understanding of the materials!

Introduction to Projections

First, let's talk about "**projections**". Projection is a linear transformation from a vectorspace to itself. Let's take a look at a simple example. In 2D plane, you have two vectors "**a**" and "**b**". You want to know the vector "**b**" that's projected on to the vector "**a**". Let's call this projected vector "**p**". Since "**p**" is projected on to "**a**", using some constant coefficient "**x**", we can represent "**p**" respect to "**a**". Also, we can define another vector "**e**" which is orthogonal to "**a**". If you write down all the relationships, it's something like this.



If you don't know or remember why the 3rd equation is like that, check the Part 3 for orthogonality!

OK, so the first goal here is to figure out “x” so that using the 1st equation, you can represent “p” as something times “a”. Try taking sometime solving the equations.

Here's the answer.

From ① & ②

$$e = b - x a \dots ④$$

From ③ & ④

$$a^T (b - x a) = 0$$

↓

$$a^T b - x a^T a = 0$$

↓

$$x = \frac{a^T b}{a^T a}$$

Now you know “x”, so put this in the 1st equation to see what “p” is. Since “x” is a constant, we could tweak it a little to modify the equation so that “p” is represented by “something” times the vector “b”.

So now we have this “**Projection Matrix**”. It might be difficult to fully understand from this equation form, so let me give you a concrete example below:

Let's say you have two vectors.

$$P = x a$$

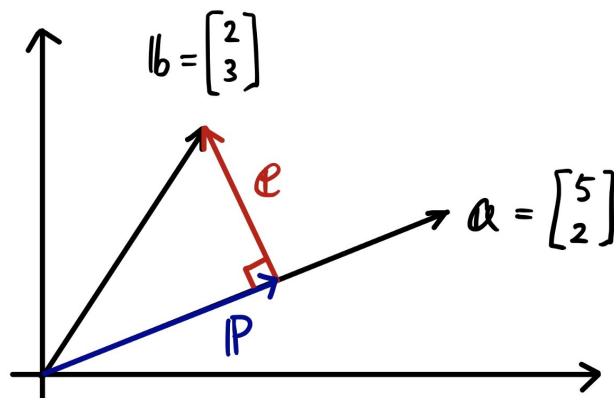
$$= \frac{a^T b}{a^T a} a$$

This part is a Constant.

$$P = a \frac{a^T b}{a^T a}$$

$$= \boxed{\frac{a a^T}{a^T a}} b$$

Projection Matrix



Just like what we did before when deriving the equation for projection, we will do the same with actual vectors.

$$P = \frac{a a^T}{a^T a} b$$

$$\begin{cases} a^T a = [5 \ 2] \begin{bmatrix} 5 \\ 2 \end{bmatrix} = 25 + 4 = 29 \\ a a^T = \begin{bmatrix} 5 \\ 2 \end{bmatrix} [5 \ 2] = \begin{bmatrix} 25 & 10 \\ 10 & 4 \end{bmatrix} \end{cases}$$

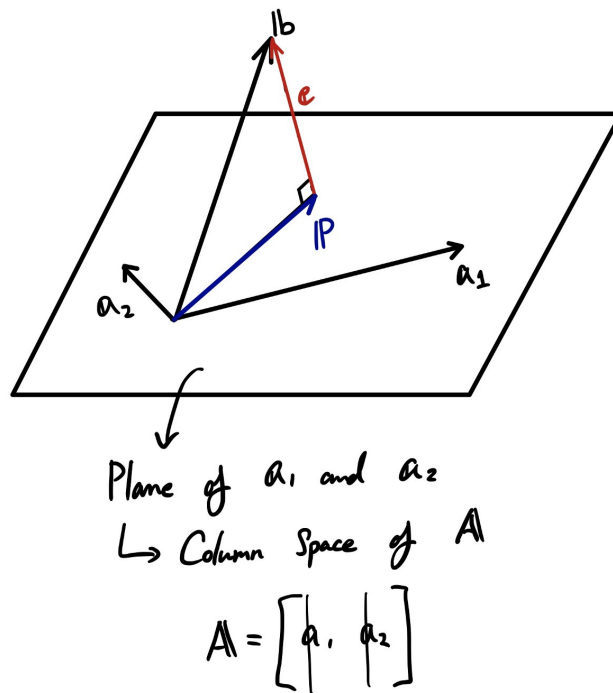
$$\therefore \frac{a a^T}{a^T a} = \frac{1}{29} \begin{bmatrix} 25 & 10 \\ 10 & 4 \end{bmatrix}$$

Now using this calculated projection matrix, we can figure out the projected vector “**p**”.

$$\begin{aligned} P &= \frac{a a^T}{a^T a} b = \frac{1}{29} \begin{bmatrix} 25 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= \frac{1}{29} \begin{bmatrix} 50 + 30 \\ 20 + 12 \end{bmatrix} \\ &= \frac{1}{29} \begin{bmatrix} 80 \\ 32 \end{bmatrix} = \frac{16}{29} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \\ &\quad \underbrace{\hspace{1.5cm}}_{\text{“a”}} \end{aligned}$$

Now you can see that the projected vector “**p**” is nothing but the scaled version of a vector “**a**”. Remember that $a = \text{transpose}[5, 2]$? It makes sense because the projected vector is on the vector “**a**”.

Now let’s take a look at a 3D example.



The plane you see there is a column space represented by column vectors “ a_1 ” and “ a_2 ”. “ A ” is a matrix that has those column vectors in there. Now what we want to do is to project vector “ b ”, which is not on this plane, to the plane.

Just like what we did in the 2D example, we define two vectors “ p ” and “ e ”.

$$\begin{cases} p = Ax = x_1 a_1 + x_2 a_2 \\ e = b - p = b - Ax \end{cases}$$

Since “ e ” is perpendicular to the plane, we can create two equations as follows. By combining those two equations and modifying it accordingly, we get the scaling factor “ x ”.

e is Perpendicular to the plane.

$$\hookrightarrow \begin{cases} a_1^T e = 0 \\ a_2^T e = 0 \end{cases}$$

$\downarrow$

$$\begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix} (b - Ax) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\downarrow$

$$A^T (b - Ax) = 0$$

$\downarrow$

$$A^T A x = A^T b$$

$$\therefore \boxed{x = (A^T A)^{-1} A^T b}$$

Using the above scaling factor, we can get "p".

$$\text{Since } p = Ax$$

$$= A \underbrace{(A^T A)^{-1} A^T}_{II} b$$

Projection Matrix P

Now we get the "**Projection Matrix**"! This time, it's a more general form so that it could be used for dimensions more than 3.

Let's take a look at the properties.

Properties

• $P^T = P$

↳ Proof

$$\begin{aligned} P &= A(A^T A)^{-1} A^T \\ P^T &= A(A^T A)^{-T} A^T \\ &= A(A^T A)^{-1} A^T \end{aligned}$$

$(XY)^T = Y^T X^T$
Taking Transpose Inside

• $P^2 = P$

↳ Proof

$$\begin{aligned} P^2 &= A(A^T A)^{-1} \underbrace{A^T A}_{(A^T A)} (A^T A)^{-1} A^T \\ &= A(A^T A)^{-1} A^T \\ &= P \end{aligned}$$

$(A^T A)^{-1} (A^T A) = I$ (Identity Matrix)

This projection part is very important so make sure you understand it well! It's used to derive the formula for many of the algorithms used in Machine Learning. To give you an example, a well know machine learning technique called "Support Vector Machine (SVM)" when calculating the margin. If you know what I'm talking about, great! If not, that's ok. I will cover this topic as well sometime in the future.

Projection (linear algebra) - Wikipedia

In linear algebra and functional analysis, a projection is a linear transformation P from a vector space to itself such...

en.wikipedia.org

Introduction to Orthonormal vectors

Now another new topic called "Orthonormal vectors". It's very simple. If two of the vectors are:

1. Unit vectors
2. Orthogonal to each other

Then they are orthonormal vectors. In equations, it's expressed as something like the following:

Orthonormal Vectors

$$q_i^T q_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

We covered orthogonality in Part 3. It's basically saying that two of the vectors are perpendicular to each other, or the dot product is zero.

Here, I'm going to introduce you to the unit vector. It's just normalizing the vector so that the length of the unit vector is always 1.

Unit Vector

$$\hat{u} = \frac{u}{\|u\|} \quad (\|u\|: \text{Norm of } u)$$

As an example, if you have a vector like the one below, the unit vector can be calculated accordingly.

example $u = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

$$\|u\| = \sqrt{(1)^2 + (-1)^2 + (2)^2} = \sqrt{1 + 1 + 4} = \sqrt{6}$$

$$\hat{u} = \frac{u}{\|u\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Unit vector - Wikipedia

By definition, in a Euclidean space the dot product of two unit vectors is a scalar value amounting to the cosine of...

en.wikipedia.org

So just to summarize, orthonormal vectors are vectors that are orthogonal to each other and are also unit vectors.

Orthonormality - Wikipedia

In linear algebra, two vectors in an inner product space are orthonormal if they are orthogonal and unit vectors. A set...

en.wikipedia.org

Introduction to Orthogonal matrix

“**Orthogonal Matrix**” is nothing but the square matrix whose rows and columns are orthonormal vectors.

Orthogonal Matrix

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix}$$

$$Q \in \mathbb{R}^{n \times n} \quad (\text{Square Matrix})$$

Orthogonal matrix has an interesting property that multiplied by its own transpose, the result becomes an identity matrix. Therefore, we can say that the transpose of the orthogonal matrix is equal to the inverse of the orthogonal matrix. This property is very important when it comes to eigen decomposition and singular value decomposition (covered in the later stories) so keep that in mind!

$$Q^T Q = I \rightarrow Q^T = Q^{-1}$$

↳ Proof

$$\begin{aligned} Q^T Q &= \begin{bmatrix} \overline{q_1} \\ \overline{q_2} \\ \vdots \\ \overline{q_n} \end{bmatrix} \begin{bmatrix} | & | & \cdots & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & \cdots & | \end{bmatrix} \\ &= \begin{bmatrix} q_1 q_1 & q_1 q_2 & \cdots & q_1 q_n \\ q_2 q_1 & q_2 q_2 & \cdots & q_2 q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n q_1 & q_n q_2 & \cdots & q_n q_n \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \\ &= I \quad (\text{Identity Matrix}) \end{aligned}$$

Remember that $q_i q_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$

Above is just showing what multiplying with its own transpose makes it an identity matrix.

Orthogonal matrix - Wikipedia

An orthogonal matrix Q is necessarily invertible (with inverse Q^T), unitary ($Q^H = Q^T$) and therefore normal ($Q^T Q = Q Q^T = I$) in the reals. The...

en.wikipedia.org

Introduction to Gram-Schmidt process

So, let's say you have a bunch of vectors, but they are not orthonormal vectors. You want to turn them into orthonormal vectors, but how can we do that? This is where “**Gram-Schmidt Process**” comes into play.

Gram-Schmidt Process

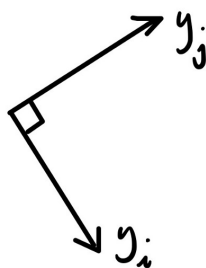
Original Vectors : x_1, x_2, x_3



Orthonormal Vectors : y_1, y_2, y_3

So that

$$y_i y_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$



Let's take a step by step example of how the process works. The 1st step is very simple. You just assign your vector to be “ y_1 ”. Later, you are going to convert all these y 's to unit vectors, but first let's deal with the orthogonality.

Step 1

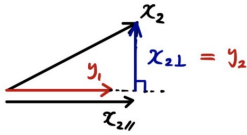
$$y_1 = x_1$$



From Step 2, this is where the actual process comes into play. So we want the outcome vectors to be orthogonal to each other. Look at the triangle plot below. We already defined “ $\mathbf{y}_1$ ” and now we want the vector “ $\mathbf{y}_2$ ” that’s orthogonal to “ $\mathbf{y}_1$ ”, but derived from the original vector “ $\mathbf{x}_2$ ”. If we decompose “ $\mathbf{x}_2$ ” to parallel and perpendicular vectors, the only vector that we want in the end is the one that’s perpendicular. That’s why we are subtracting the parallel version of the “ $\mathbf{x}_2$ ” from the original vector “ $\mathbf{x}_2$ ”.

Step 2

$$\mathbf{y}_2 = \mathbf{x}_2 - \mathbf{x}_{2//}$$



$$\begin{cases} \mathbf{x}_{2//} = a \mathbf{y}_1 = a \mathbf{x}_1 \dots \textcircled{1} \\ \mathbf{x}_2 - \mathbf{x}_{2//} = \mathbf{x}_{2\perp} \dots \textcircled{2} \\ \mathbf{y}_1^T \cdot \mathbf{x}_{2\perp} = 0 \rightarrow \mathbf{x}_1^T \cdot \mathbf{x}_{2\perp} = 0 \dots \textcircled{3} \end{cases}$$

From $\textcircled{2}$ & $\textcircled{3}$

$$\mathbf{x}_1^T (\mathbf{x}_2 - \mathbf{x}_{2//}) = 0$$

Insert $\textcircled{1}$

$$\mathbf{x}_1^T (\mathbf{x}_2 - a \mathbf{x}_1) = 0$$

$$\mathbf{x}_1^T \mathbf{x}_2 - a \mathbf{x}_1^T \mathbf{x}_1 = 0$$

$$\therefore a = \frac{\mathbf{x}_1^T \mathbf{x}_2}{\mathbf{x}_1^T \mathbf{x}_1}$$

Insert this to $\textcircled{1}$

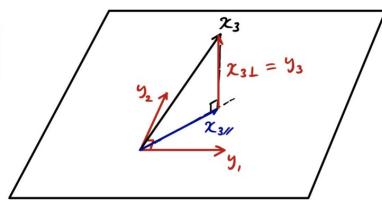
$$\mathbf{x}_{2//} = \frac{\mathbf{x}_1^T \mathbf{x}_2}{\mathbf{x}_1^T \mathbf{x}_1} \mathbf{x}_1$$

$$\mathbf{y}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_1^T \mathbf{x}_2}{\mathbf{x}_1^T \mathbf{x}_1} \mathbf{x}_1$$

Let’s take a look at the 3rd step.

Step 3

$$\mathbf{y}_3 = \mathbf{x}_3 - \mathbf{x}_{3//}$$



$$\mathbf{y}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_2^T \mathbf{x}_3}{\mathbf{x}_2^T \mathbf{x}_2} \mathbf{x}_2 - \frac{\mathbf{x}_1^T \mathbf{x}_3}{\mathbf{x}_1^T \mathbf{x}_1} \mathbf{x}_1$$

By now, you might have noticed the patterns here.

If we generalize the pattern, the Gram-Schmidt Process is described as follows.

In General

$$y_k = x_k - \sum_{j=1}^{k-1} \text{proj}(x_j)$$

$$\text{proj}(x_j) = \frac{x_j^T x_k}{x_j^T x_j} x_j$$

In the end, don't forget to normalize to unit vectors!

Summary

Projections

Projection is a linear transformation projecting the vector or space to some other vectorspace

Orthonormal vectors

Orthonormal vectors are vectors that are 1) orthogonal to each other & 2) unit vectors

Orthogonal matrix

Orthogonal matrix is a matrix with orthonormal vectors in rows and columns

Gram Schmidt process

Gram Schmidt Process is a process to convert vectors into orthonormal vectors.

I hope this helps! See you next time!